

Junseo Lee

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Research Interests

Computer Systems, ML Systems, Compilers & Runtime Systems, Autotuning, Hardware-Aware Optimization

Education

Seoul National University Master of Data Science <i>Thesis: Towards Batched Activation Sparsity in LLM Decoding</i> Computing Optimization and Data-driven Exploration (CODE) Lab	Mar 2024 – Aug 2026 Advisor: Hyojin Sung GPA 3.52/4.0
Yonsei University B.A. in Applied Statistics	Mar 2020 – Feb 2024 GPA 3.92/4.3
University of California, Berkeley UCEAP Exchange Program, College of Letters & Science	Aug 2022 – May 2023

Research Experience

Computing Optimization and Data-driven Exploration (CODE) Lab – SNU <i>Research Collaborator</i> <i>Graduate Researcher</i>	Sep 2026 – Present Sep 2024 – Aug 2026 Advisor: Hyojin Sung
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Autotuning Framework for Deep Learning Compilers Project

HYPERF (HPDC 2025)

- Evaluated the HYPERF autotuning framework against an OpenTuner baseline on PolyBench kernels, achieving a 6.0× average execution speedup and 1.2× faster tuning convergence.

SpMV Workload Optimization (HYPERF Follow-up; SC 2026 submission)

- Built a performance dataset for SpMV row-binning across matrices from the SuiteSparse Matrix Collection to characterize when binning improves load balance and performance.
- Developed a hierarchical ML-based selector using XGBoost to identify when to apply row-binning and C5.0 decision tree to select promising configurations.
- Integrated the predicted top configurations into HYPERF v2's autotuning sampler, reducing tuning trials by 60% (570 → 230) while reaching within 5% of the best-performing configuration.

Autotuning Sampler and TVM Cost Model (In Preparation for OSDI)

- Adapting a bandit-based Rising-Successive Rejects (R-SR) strategy to

HYPERF v2's autotuning sampler for budget allocation across a hierarchical search space spanning algorithmic choices, compiler transformations, and scheduling parameters.

- Applying bootstrap aggregation to TVM's XGBoost cost model to improve robustness under limited training data.

Towards Batched Activation Sparsity in LLM Decoding (Master's thesis)

- Investigated whether blocked activation sparsity can be shared across sequences in batched autoregressive LLM decoding without compromising model quality.
- Developed an oracle analysis and Triton block-sparse GEMM kernels to characterize achievable sparsity and kernel-level speedups under batching.
- Found substantial per-decoding-step sparsity potential (64% at batch size 16), while revealing a significant gap between the oracle upper bound and practical online sparsification due to quality degradation and autoregressive error accumulation.

Publications

Park, J.*, Shin Y.*, Lee, J., **Lee, J.**, Kim, J., Kwon, O., & Sung, H. (*: co-first authors)
HYPERF: End-to-End Autotuning Framework for High-Performance Computing
Proceedings of the International Symposium on High-Performance Parallel and Distributed Computing (HPDC), 2025.

Teaching & Mentoring

Undergraduate Data Science Project Tutor — SNU **Sep 2024 – Dec 2024**

- Mentored 30+ undergraduate students in Python programming and data science projects as part of a 4-person teaching team.

Skills

Programming: Python, C/C++, CUDA, Triton, Bash, R

Systems & Tools: TVM, NVIDIA Nsight Systems/Compute, OpenMP, MPI, Slurm, Docker, Git

Languages

Korean (native), English (professional), Spanish (basic)